GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
tRNA Methylation (GO:0030488)	0.25364110	34	3.150e−07	1.692e−03	TARBP1:75 TRMT9B:459 TRMT1L:460 LCMT2:474 TRMT10A:503 TRMT5:538
Fatty Acid Catabolic Process (GO:0009062	0.18065022	55	3.695e−06	9.922e−03	EHHADH:78 ACAA2:83 ACADM:92 ACAD11:106 ECI2:180 HADHA:245
Chemical Synaptic Transmission (GO:00072	−0.09386238	188	1.008e−05	1.353e−02	GABRR1:16 STXBP1:47 GABRB3:122 RIMBP2:128 RPS6KA2:138 CHRFAM7A:139
Regulation Of Synaptic Transmission, Glu	−0.20396950	40	8.215e−06	1.353e−02	DGKI:24 PTK2B:63 MAPK8IP2:109 SHANK3:146 CACNG3:159 SYT1:199
Alpha−Amino Acid Catabolic Process (GO:1	0.22897878	29	1.998e−05	1.789e−02	ALDH4A1:80 TAT:187 HIBCH:219 KMO:432 GOT2:558 ACMSD:791
Mitochondrial Gene Expression (GO:014005	0.13059664	91	1.743e−05	1.789e−02	PTCD3:67 TEFM:95 MRPS9:243 MRPL43:445 MRPS18A:504 MRPS22:599
NADH Dehydrogenase Complex Assembly (GO:	0.18870308	41	2.957e−05	1.925e−02	ECSIT:79 FOXRED1:98 NDUFA9:169 NDUFB8:308 DMAC2:347 TMEM126B:499
RNA Methylation (GO:0001510)	0.17131883	48	4.105e−05	1.925e−02	TARBP1:75 TRMT9B:459 LCMT2:474 RBM15B:480 TRMT10A:503 TFB2M:604
Fatty Acid Beta−Oxidation (GO:0006635)	0.17779355	45	3.757e−05	1.925e−02	EHHADH:78 ACAA2:83 ACADM:92 ACAD11:106 ACOX2:174 ECI2:180
Mitochondrial Respiratory Chain Complex	0.18870308	41	2.957e−05	1.925e−02	ECSIT:79 FOXRED1:98 NDUFA9:169 NDUFB8:308 DMAC2:347 TMEM126B:499
Non−Motile Cilium Assembly (GO:1905515)	0.22186584	29	3.587e−05	1.925e−02	CEP89:210 ARL13B:282 CEP250:666 GORAB:887 ENKD1:903 C2CD3:1040
Steroid Metabolic Process (GO:0008202)	0.15825673	56	4.301e−05	1.925e−02	CYP2R1:21 CYP1A1:72 CYP26C1:138 DHRS4:194 CYP17A1:225 NPC1:277
Organic Hydroxy Compound Transport (GO:0	0.19316430	36	6.141e−05	2.537e−02	ABCG5:141 SLC51A:211 SLC51B:334 RALBP1:404 STRA6:466 SLCO1A2:549
Cilium Organization (GO:0044782)	0.08572650	183	6.897e−05	2.646e−02	TCTN1:34 CCDC66:71 CFAP43:116 TCTN2:182 CEP89:210 IQUB:251
Mitochondrial Translation (GO:0032543)	0.12107935	87	9.829e−05	3.519e−02	PTCD3:67 MRPS9:243 MTIF2:269 MRPL43:445 GFM1:452 MRPS18A:504
Monocarboxylic Acid Transport (GO:001571	0.15778679	50	1.157e−04	3.656e−02	SLC5A12:5 SLC16A5:183 SLC22A13:205 SLC51A:211 SLC51B:334 SLC6A12:341
tRNA Modification (GO:0006400)	0.14344262	61	1.094e−04	3.656e−02	TARBP1:75 TRMT9B:459 LCMT2:474 TRMT10A:503 TRMT1:643 NSUN6:657
Mitochondrial Respiratory Chain Complex	0.13923841	62	1.532e−04	4.167e−02	ECSIT:79 FOXRED1:98 NDUFA9:169 NDUFB8:308 DMAC2:347 TMEM126B:499
Modulation Of Chemical Synaptic Transmis	−0.11663539	89	1.480e−04	4.167e−02	DGKI:24 TRIO:77 MAPK8IP2:109 PTEN:151 GRID1:172 BAIAP3:176
Positive Regulation Of Synaptic Transmis	−0.15330835	51	1.552e−04	4.167e−02	PTK2B:63 SHANK3:146 CACNG3:159 BAIAP3:176 CRHR2:186 SHANK2:188
Protein Glycosylation (GO:0006486)	0.10449613	107	1.961e−04	5.016e−02	TET1:7 B3GNT4:247 GALNT5:250 POGLUT1:257 NPC1:277 FKTN:291
Double−Strand Break Repair (GO:0006302)	0.09222659	134	2.397e−04	5.852e−02	ANKLE1:168 WRN:207 ERCC5:259 NPC1:277 RNF168:526 PRKDC:671
Cell Junction Assembly (GO:0034329)	−0.13319888	63	2.617e−04	6.112e−02	SDK1:80 TLN1:94 SHANK3:146 PTEN:151 SHANK2:188 DBNL:295
Cilium Assembly (GO:0060271)	0.07535552	197	2.867e−04	6.333e−02	TCTN1:34 CCDC66:71 CFAP43:116 TCTN2:182 CCDC39:197 CEP89:210
Nervous System Development (GO:0007399)	−0.06116961	301	2.948e−04	6.333e−02	VAX2:8 ADGRB1:27 SH3GL3:43 SCIN:60 CNTN4:70 SDK1:80
Monocarboxylic Acid Catabolic Process (G	0.24581269	18	3.071e−04	6.344e−02	CYP26C1:138 AGXT:626 AGXT2:1549 HOGA1:1721 IDUA:1865 LPIN3:1890
Muscle Contraction (GO:0006936)	−0.12129799	72	3.819e−04	7.598e−02	MYH13:5 TACR2:20 MYH2:116 LMOD1:267 MYH8:292 EMD:399
Anterograde Trans−Synaptic Signaling (GO	−0.08793602	134	4.616e−04	8.717e−02	GABRR1:16 GABRB3:122 RPS6KA2:138 SYT1:199 KCNQ3:245 HRH3:326
Signal Release From Synapse (GO:0099643)	−0.18773908	29	4.707e−04	8.717e−02	STXBP1:47 SYT1:199 SLC32A1:218 HRH3:326 CPLX1:346 NRXN2:541
Fatty Acid Oxidation (GO:0019395)	0.15246630	43	5.489e−04	9.828e−02	EHHADH:78 ACAA2:83 ACADM:92 ACAD11:106 ECI2:180 HADHA:245
Bile Acid And Bile Salt Transport (GO:00	0.24608941	16	6.568e−04	1.080e−01	SLC51A:211 SLC51B:334 SLCO1A2:549 SLC10A2:873 ATP8B1:1063 SLCO1B1:1345
Central Nervous System Development (GO:0	−0.06807124	213	6.638e−04	1.080e−01	VAX2:8 SH3GL3:43 SCIN:60 CNTN4:70 PTEN:151 SHROOM2:168
Double−Strand Break Repair Via Homologou	0.10413484	90	6.581e−04	1.080e−01	FANCM:81 WRN:207 ERCC5:259 NPC1:277 RNF169:712 RAD54B:807
Glutamate Receptor Signaling Pathway (GO	−0.19194577	26	7.097e−04	1.100e−01	PTK2B:63 HOMER2:325 TRPM1:356 GRM5:467 GRM6:538 GRIN2A:593
Recombinational Repair (GO:0000725)	0.11246635	76	7.165e−04	1.100e−01	WRN:207 ERCC5:259 NPC1:277 RHNO1:624 RNF169:712 RAD54B:807
Fatty Acid Metabolic Process (GO:0006631	0.09605472	103	7.817e−04	1.166e−01	ABCB11:65 CYP1A1:72 ACADM:92 CPT1C:97 PLA2G1B:146 ACOX2:174
Negative Regulation Of Fatty Acid Biosyn	0.36494792	7	8.274e−04	1.201e−01	TRIB3:315 ACADVL:362 BRCA1:932 UBR4:2343 DCAF5:2724 SIRT1:2866
Fat−Soluble Vitamin Metabolic Process (G	0.23922218	16	9.266e−04	1.214e−01	CYP2R1:21 CYP1A1:72 CYP11A1:443 GGCX:571 CYP24A1:655 LRAT:802
Nitrogen Compound Transport (GO:0071705)	0.08351818	133	9.200e−04	1.214e−01	CUBN:20 RHAG:185 SLC22A13:205 AQP9:228 RHCE:231 SLC6A6:256
Positive Regulation Of Synaptic Transmis	−0.26609819	13	8.963e−04	1.214e−01	PTK2B:63 SHANK3:146 CACNG3:159 GRIN2A:593 NPS:626 TSHZ3:724

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_DEVELOPMENTAL_BIOLOGY	−0.09071584	680	1.627e−15	1.053e−11	COL6A2:22 KRT20:37 DOCK1:38 COL6A1:45 TRIO:77 AP2A2:78
REACTOME_NERVOUS_SYSTEM_DEVELOPMENT	−0.09599440	390	1.027e−10	3.326e−07	COL6A2:22 DOCK1:38 COL6A1:45 TRIO:77 AP2A2:78 TLN1:94
REACTOME_INFECTIOUS_DISEASE	−0.07237993	556	7.552e−09	1.630e−05	ADCY1:3 GOLGA7:26 DOCK1:38 SH3GL3:43 DYNC1H1:53 ITPR1:54
HOUNKPE_HOUSEKEEPING_GENES	−0.06326255	712	1.368e−08	1.772e−05	PHRF1:32 NSA2:36 HCFC1:66 SND1:113 U2AF2:117 PTEN:151
REACTOME_NEURONAL_SYSTEM	−0.10094367	270	1.325e−08	1.772e−05	ADCY1:3 CAMK2B:6 GABRR1:16 STXBP1:47 NBEA:51 AP2A2:78
REACTOME_CELLULAR_RESPONSES_TO_STIMULI	−0.07536497	469	2.924e−08	3.156e−05	CAMK2B:6 EP400:7 DNAJA4:18 RPTOR:25 DYNC1H1:53 SRPRA:84
REACTOME_SIGNALING_BY_RECEPTOR_TYROSINE_	−0.08094527	378	7.967e−08	6.585e−05	COL6A2:22 DOCK1:38 SH3GL3:43 COL6A1:45 ITPR1:54 PTK2B:63
KIM_ALL_DISORDERS_OLIGODENDROCYTE_NUMBER	−0.07327046	464	8.135e−08	6.585e−05	CAMK2B:6 DOCK4:35 TCFL5:42 STXBP1:47 DYNC1H1:53 TMED2:71
REACTOME_SIGNALING_BY_WNT	−0.11686042	161	3.344e−07	2.349e−04	ITPR1:54 PDE6B:74 AP2A2:78 PSMA3:205 TNRC6A:221 AGO3:280
WP_MRNA_PROCESSING	−0.18062307	66	3.989e−07	2.349e−04	U2AF2:117 SNRPN:156 SNRPA:161 PTBP2:163 SF3A3:270 SFSWAP:301
BLALOCK_ALZHEIMERS_DISEASE_DN	−0.05465004	766	3.869e−07	2.349e−04	STXBP1:47 DYNC1H1:53 ITPR1:54 PTPRE:61 PTK2B:63 TMED2:71
REACTOME_PROCESSING_OF_CAPPED_INTRON_CON	−0.11198787	170	5.083e−07	2.743e−04	SLBP:73 U2AF2:117 SNRPN:156 SNRPA:161 SF3A3:270 SYMPK:283
BENPORATH_ES_WITH_H3K27ME3	−0.05465478	733	6.619e−07	3.297e−04	CAMK2B:6 VAX2:8 SORCS1:10 IGFBP3:13 SMPD3:23 DGKI:24
REACTOME_DISEASES_OF_SIGNAL_TRANSDUCTION	−0.08108163	297	1.734e−06	8.023e−04	CAMK2B:6 TLN1:94 SND1:113 HDAC4:118 PTEN:151 PSMA3:205
REACTOME_NEUROTRANSMITTER_RECEPTORS_AND_	−0.12668765	118	2.092e−06	9.031e−04	ADCY1:3 CAMK2B:6 GABRR1:16 NBEA:51 AP2A2:78 GABRB3:122
CARRILLOREIXACH_HEPATOBLASTOMA_VS_NORMAL	0.05013951	785	2.477e−06	1.003e−03	PTGFR:6 ACY1:35 IDO2:51 ABCB11:65 CYP1A1:72 HMCES:77
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	−0.36237930	14	2.677e−06	1.020e−03	EP400:7 POLE:102 MMP17:237 GALNT9:261 SFSWAP:301 ANKLE2:394
NIKOLSKY_BREAST_CANCER_8P12_P11_AMPLICON	0.21621579	38	4.029e−06	1.450e−03	ANK1:36 IDO2:51 RAB11FIP1:53 NSD3:64 DDHD2:111 IKBKB:135
REACTOME_METABOLIC_DISORDERS_OF_BIOLOGIC	0.25463672	26	7.026e−06	2.275e−03	CYP2R1:21 ACY1:35 CYP26C1:138 CYP17A1:225 CYP11A1:443 CYP24A1:655
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	−0.05052726	699	6.945e−06	2.275e−03	CAMK2B:6 IGFBP3:13 RPTOR:25 SLBP:73 ZNF704:86 USP2:89
WP_2Q37_COPY_NUMBER_VARIATION_SYNDROME	−0.13948118	86	8.004e−06	2.468e−03	KLHL30:21 KIF1A:34 GYS1:68 HDAC4:118 TRAF3IP1:131 GPC1:177
KIM_ALL_DISORDERS_CALB1_CORR_UP	−0.07010637	337	1.097e−05	3.088e−03	CAMK2B:6 DOCK4:35 STXBP1:47 DYNC1H1:53 ITPR1:54 PTK2B:63
KEGG_VALINE_LEUCINE_AND_ISOLEUCINE_DEGRA	0.19407544	43	1.079e−05	3.088e−03	EHHADH:78 ACAA2:83 ACADM:92 ACAD8:189 HIBCH:219 HADHA:245
REACTOME_FATTY_ACID_METABOLISM	0.10783639	139	1.191e−05	3.213e−03	CYP1A1:72 EHHADH:78 ACAA2:83 ACADM:92 ACAD11:106 ACOT12:109
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	−0.12650118	100	1.275e−05	3.304e−03	CABLES2:31 TCFL5:42 ZBTB46:201 RAE1:312 HRH3:326 MYT1:327
LIM_MAMMARY_STEM_CELL_UP	−0.06686132	364	1.340e−05	3.311e−03	SORCS1:10 UPP1:12 IGFBP3:13 PTPRE:61 TRIM29:81 RNF165:85
REACTOME_SIGNALING_BY_GPCR	−0.05637608	517	1.380e−05	3.311e−03	ADCY1:3 CAMK2B:6 TACR2:20 DGKI:24 GPR55:29 GAL:40
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	−0.06088005	435	1.555e−05	3.597e−03	CAMK2B:6 ADGRB1:27 GAL:40 CNTN4:70 MAPK8IP2:109 OTUD7A:114
BENPORATH_SUZ12_TARGETS	−0.04873091	684	1.774e−05	3.844e−03	VAX2:8 SORCS1:10 DGKI:24 HSF4:30 REPS2:41 CRABP1:57
BENPORATH_EED_TARGETS	−0.04902705	675	1.781e−05	3.844e−03	CAMK2B:6 VAX2:8 SORCS1:10 IGFBP3:13 DGKI:24 HSF4:30
REACTOME_INFLUENZA_INFECTION	−0.13931432	79	1.906e−05	3.982e−03	RAE1:312 RPL7A:408 RPS8:445 NUP93:463 NUP37:518 KPNA3:564
REACTOME_BIOLOGICAL_OXIDATIONS	0.11162574	123	1.981e−05	4.008e−03	CYP2R1:21 ACY1:35 ALDH1A1:60 CYP1A1:72 CYP26C1:138 MGST3:215
REACTOME_TRANSMISSION_ACROSS_CHEMICAL_SY	−0.09439264	170	2.296e−05	4.506e−03	ADCY1:3 CAMK2B:6 GABRR1:16 STXBP1:47 NBEA:51 AP2A2:78
WP_MALIGNANT_PLEURAL_MESOTHELIOMA	−0.07047266	301	2.869e−05	5.465e−03	RPTOR:25 TRAF2:55 HCFC1:66 PTEN:151 PTK2:171 FZD10:198
LEIN_NEURON_MARKERS	−0.17006496	47	5.554e−05	1.028e−02	JAKMIP1:1 CAMK2B:6 REPS2:41 SNRPN:156 SYT1:199 DPP6:200
REACTOME_HIV_INFECTION	−0.09758035	142	6.206e−05	1.032e−02	AP2A2:78 PSMA3:205 RAE1:312 NUP93:463 FURIN:516 NUP37:518
REACTOME_GENE_SILENCING_BY_RNA	−0.13944262	69	6.304e−05	1.032e−02	MOV10L1:115 TNRC6A:221 AGO3:280 RAE1:312 NUP93:463 NUP37:518
REACTOME_TRANSPORT_OF_MATURE_TRANSCRIPT_	−0.15351799	57	6.185e−05	1.032e−02	SLBP:73 U2AF2:117 SYMPK:283 RAE1:312 SLU7:380 NUP93:463
WONG_MITOCHONDRIA_GENE_MODULE	0.09467620	151	6.191e−05	1.032e−02	CHDH:32 CRYZ:70 NDUFA9:169 SLC16A5:183 ATP13A3:249 GCDH:286
KEGG_SPLICEOSOME	−0.15742537	54	6.374e−05	1.032e−02	U2AF2:117 SNRPA:161 SF3A3:270 SMNDC1:331 SRSF10:354 SLU7:380

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Mitochondrial Diseases	0.07320435	308	1.173e−05	1.072e−01	ALB:19 TXNRD1:48 CRYZ:70 ECSIT:79 FOXRED1:98 NDUFA9:169
Schizophrenia	−0.03664206	1278	2.194e−05	1.072e−01	ADCY1:3 CAMK2B:6 GABRR1:16 DOCK4:35 TMEM132D:39 STXBP1:47
Central neuroblastoma	−0.03773073	1086	4.668e−05	1.140e−01	CAMK2B:6 IGFBP3:13 KIF1A:34 DOCK4:35 GAL:40 ITPR1:54
Neuroblastoma	−0.03760430	1113	4.095e−05	1.140e−01	CAMK2B:6 IGFBP3:13 KIF1A:34 DOCK4:35 GAL:40 ITPR1:54
HIV Infections	−0.05230156	515	6.165e−05	1.204e−01	KRT20:37 TMEM132D:39 PTK2B:63 TMED2:71 VIPR1:103 PLEC:142
Cholestasis	0.07462788	229	1.098e−04	1.532e−01	ABCB11:65 SLC23A1:68 ABCG5:141 PLA2G1B:146 SLC51A:211 CYP17A1:225
Neurodevelopmental Disorders	−0.09339233	146	1.041e−04	1.532e−01	STXBP1:47 ITPR1:54 HCFC1:66 CNTN4:70 TRIO:77 GABRB3:122
Renal salt wasting	0.29337535	14	1.448e−04	1.768e−01	KCNJ1:363 SCNN1A:442 CYP11A1:443 SARS2:547 SCNN1B:756 CYP21A2:1154
Anemia, hereditary spherocytic hemolytic	0.46520717	5	3.153e−04	2.093e−01	ANK1:36 SPTA1:100 SLC4A1:221 EPB42:370 SPTB:1502 NA
Ciliopathies	0.08562931	150	3.110e−04	2.093e−01	TCTN1:34 TCTN2:182 CCDC39:197 NPHP3:252 NME8:275 ARL13B:282
Downturned corners of mouth	−0.13956583	57	2.729e−04	2.093e−01	HERC2:111 HDAC4:118 SNRPN:156 SMC3:196 CPLX1:346 SMC1A:383
Hyponatremia	0.18387534	32	3.214e−04	2.093e−01	CLDN16:66 SCNN1A:442 CYP11A1:443 COMT:449 SARS2:547 MC2R:738
Liver carcinoma	−0.02565623	2215	2.144e−04	2.093e−01	EP400:7 IGFBP3:13 SMPD3:23 GAL:40 TCFL5:42 BNC2:50
Nephrocalcinosis	0.15779439	45	2.537e−04	2.093e−01	TRPM6:4 CYP2R1:21 CLDN16:66 SLC4A1:221 KCNJ1:363 GRHPR:385
Spherocytosis	0.46520717	5	3.153e−04	2.093e−01	ANK1:36 SPTA1:100 SLC4A1:221 EPB42:370 SPTB:1502 NA
Night blindness, congenital stationary	−0.21906490	21	5.131e−04	2.785e−01	PDE6B:74 USP11:91 GPR34:345 TRPM1:356 CABP4:454 GRM6:538
Autism Spectrum Disorders	−0.05360059	369	4.625e−04	2.785e−01	JAKMIP1:1 NBEA:51 ITPR1:54 CNTN4:70 TRIO:77 HERC2:111
Congenital cerebral hernia	0.16339271	38	4.969e−04	2.785e−01	TCTN2:182 FKTN:291 TMEM67:446 WDPCP:515 ARHGAP31:779 LRAT:802
Type 1 muscle fiber predominance	−0.36933337	7	7.152e−04	3.677e−01	COL6A2:22 COL6A1:45 NEB:59 TNNT1:406 MYH7:705 COL6A3:5124.5
Primary hyperoxaluria, type I	0.27900575	12	8.200e−04	4.005e−01	GRHPR:385 AGXT:626 DAO:1605 AGT:1648 PTHLH:1692 PRDX5:1771
Neoplasm Metastasis	−0.02211344	2446	9.539e−04	4.235e−01	UPP1:12 IGFBP3:13 COL6A2:22 SMPD3:23 ADGRB1:27 DOCK4:35
Undifferentiated carcinoma	−0.07649316	158	9.507e−04	4.235e−01	PTEN:151 SYT1:199 IL18:258 MYC:324 NSMF:365 KRT71:366
Cholestasis in newborn	0.10520772	80	1.167e−03	4.768e−01	ABCB11:65 EHHADH:78 ABCG5:141 HADHA:245 NPHP3:252 NPC1:277
Fragile skin	−0.20972631	20	1.172e−03	4.768e−01	PLEC:142 ITGB4:388 PKP1:465 DSP:638 COL5A1:669 ITGA3:741
Aneurysm, Ruptured	−0.34825540	7	1.420e−03	4.911e−01	RNH1:106 PPP1R13L:504 SHC3:514 NOS3:893 ACE:975 FN1:5124.5
Round, full face	−0.12297485	56	1.478e−03	4.911e−01	COL6A2:22 COL6A1:45 HDAC4:118 SEMA5A:133 PTEN:151 SH3BP2:251
Anaplastic Ependymoma	−0.22918710	16	1.508e−03	4.911e−01	IGFBP3:13 PTEN:151 SMARCB1:621 NOTCH1:672 TIMP3:921 FOXJ1:998
Glioma	−0.02666157	1392	1.363e−03	4.911e−01	CAMK2B:6 IGFBP3:13 RPTOR:25 ADGRB1:27 DOCK1:38 GAL:40
Polyuria	0.18146915	26	1.369e−03	4.911e−01	CLDN16:66 NPHP3:252 KCNJ1:363 TMEM67:446 SLC12A3:546 SARS2:547
Round face	−0.12297485	56	1.478e−03	4.911e−01	COL6A2:22 COL6A1:45 HDAC4:118 SEMA5A:133 PTEN:151 SH3BP2:251
Hyperkalemia	0.23530738	15	1.608e−03	5.066e−01	CYP17A1:225 SCNN1A:442 CYP11A1:443 INVS:697 SCNN1B:756 WNK4:1128
Metabolic acidosis	0.11553897	62	1.679e−03	5.126e−01	EHHADH:78 ACADM:92 LCT:133 SLC4A1:221 GCDH:286 SLC9A4:403
EMG: neuropathic changes	−0.23105860	15	1.951e−03	5.171e−01	DYNC1H1:53 NEB:59 HSPB3:405 VRK1:530 CAPN3:552 MYH7:705
Glioblastoma	−0.02683034	1258	2.024e−03	5.171e−01	IGFBP3:13 DGKI:24 ADGRB1:27 DOCK1:38 GAL:40 TCFL5:42
Lethargy	0.09449603	90	1.991e−03	5.171e−01	ACADM:92 FOXRED1:98 SLC4A1:221 ACADSB:293 CPT2:314 ACADVL:362
Mammary Neoplasms	−0.02441243	1578	2.010e−03	5.171e−01	CAMK2B:6 IGFBP3:13 SMPD3:23 KRT20:37 DOCK1:38 GAL:40
Purpura	0.14454806	38	2.064e−03	5.171e−01	ITGA2B:409 GP1BB:476 GP9:825 MEFV:938 NFKB2:1013 PRKCD:1018
Severe myopia	−0.10261303	77	1.888e−03	5.171e−01	IGFBP3:13 PDE6B:74 OPN1LW:236 TRPM1:356 VIPR2:423 CABP4:454
Sprains and Strains	−0.39953962	5	1.975e−03	5.171e−01	PTEN:151 IL18:258 XIAP:286 TIAM1:441 MTOR:5124.5 NA
Pain, Postoperative	0.20903077	18	2.146e−03	5.218e−01	COMT:449 TAOK3:701 AP3B1:1566 TNFRSF1B:1852 CPOX:1981 PTGS2:1988

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0009642 abnormal blood homeostasis	0.03811931	1034	2.140e−04	1.019e−01	TRPM6:4 INPP5K:8 ALB:19 CLDN16:66 SLC23A1:68 CYP1A1:72
MP0000598 abnormal liver morphology	0.04780170	391	1.796e−03	1.090e−01	NAGLU:10 MAN2A1:18 ANK1:36 CYBRD1:39 ABCB11:65 CYP1A1:72
MP0005636 abnormal mineral homeostasis	0.08595584	118	1.425e−03	1.090e−01	TRPM6:4 ALB:19 ANK1:36 CYBRD1:39 CLDN16:66 GUCA2B:123
MP0003868 abnormal feces composition	0.20625437	24	4.828e−04	1.090e−01	ALB:19 PNLIP:89 CLPS:381 ABCB4:622 CYP7A1:672 SLC26A3:829
MP0009643 abnormal urine homeostasis	0.06627583	200	1.523e−03	1.090e−01	MAN2A1:18 ALB:19 CUBN:20 SLC14A2:59 CLDN16:66 SLC23A1:68
MP0008875 abnormal xenobiotic pharmacok	0.25270125	13	1.627e−03	1.090e−01	CYP1A1:72 MTF1:140 COMT:449 AHR:998 CYP1A2:1105 AS3MT:1940
MP0003252 abnormal bile duct	0.19937689	20	2.062e−03	1.090e−01	ABCB11:65 ABCG5:141 PKHD1:521 PKD2:621 ABCB4:622 P2RY13:848
MP0001666 abnormal nutrient absorption	0.14671484	40	1.379e−03	1.090e−01	ALB:19 SLC23A1:68 PNLIP:89 ABCG5:141 PLA2G1B:146 VDR:359
MP0003635 abnormal synaptic transmissio	−0.05237128	318	1.850e−03	1.090e−01	ADCY1:3 CAMK2B:6 GAL:40 STXBP1:47 NBEA:51 ITPR1:54
MP0005365 abnormal bile salt	0.16917425	26	2.893e−03	1.377e−01	ALB:19 ABCB11:65 EHHADH:78 PNLIP:89 GPBAR1:321 CYP7A1:672
MP0005085 abnormal gallbladder physiolo	0.17562582	23	3.619e−03	1.566e−01	ABCB11:65 ABCG5:141 GPBAR1:321 ABCB4:622 CYP7A1:672 CYP27A1:750
MP0000609 abnormal liver physiology	0.04711727	322	4.846e−03	1.739e−01	MAN2A1:18 ALB:19 CYBRD1:39 ABCB11:65 CYP1A1:72 EHHADH:78
MP0005332 abnormal amino acid	0.08024471	104	5.109e−03	1.739e−01	ACADM:92 MEP1A:171 HADHA:245 SLC6A6:256 ACADVL:362 LAMP2:447
MP0002066 abnormal motor capabilities/c	−0.03081932	842	5.456e−03	1.739e−01	CAMK2B:6 VAX2:8 RPTOR:25 KIF1A:34 NBEA:51 DYNC1H1:53
MP0000678 abnormal parathyroid gland	0.20739879	15	5.480e−03	1.739e−01	CHRD:263 VDR:359 PTH:582 CYP27B1:1415 PTHLH:1692 GCM2:1870
MP0001970 abnormal pain threshold	−0.06886297	121	9.691e−03	2.435e−01	ADCY1:3 GABRR1:16 KIF1A:34 HDAC4:118 PIRT:194 CHRNA4:336
MP0002139 abnormal hepatobiliary system	0.27931126	7	1.054e−02	2.435e−01	ALDH1A1:60 ABCB4:622 FABP1:698 ABCG8:995 SLC22A1:3391 NR1H3:7610.5
MP0002136 abnormal kidney physiology	0.04862541	245	1.046e−02	2.435e−01	MAN2A1:18 ALB:19 CUBN:20 SLC14A2:59 CLDN16:66 SLC23A1:68
MP0009765 abnormal xenobiotic induced	0.08543499	77	1.007e−02	2.435e−01	CYP1A1:72 HIP1:73 MTF1:140 SLC4A1:221 VDR:359 ABCC1:564
MP0004019 abnormal vitamin homeostasis	0.12375418	34	1.279e−02	2.435e−01	CUBN:20 CLDN16:66 SLC23A1:68 VDR:359 GALNT3:554 CYP24A1:655
MP0002063 abnormal learning/memory/cond	−0.04609840	269	1.122e−02	2.435e−01	ADCY1:3 MAPK8IP2:109 MBP:206 ATP1A3:213 ZDHHC8:227 L1CAM:293
MP0002064 seizures	−0.05322673	197	1.148e−02	2.435e−01	GAL:40 ITPR1:54 HDAC4:118 GABRB3:122 PTEN:151 MBP:206
MP0002882 abnormal neuron morphology	−0.03126232	694	8.974e−03	2.435e−01	VAX2:8 LOXHD1:28 KIF1A:34 NBEA:51 DYNC1H1:53 PDE6B:74
MP0002272 abnormal nervous system	−0.05988724	152	1.201e−02	2.435e−01	GABRR1:16 GABRB3:122 PLEC:142 SLC32A1:218 KCNQ3:245 SCN1A:332
MP0009745 abnormal behavioral response	−0.05803908	161	1.232e−02	2.435e−01	UPP1:12 GAL:40 GABRB3:122 PTEN:151 MBP:206 ATP1A3:213
MP0002752 abnormal somatic nervous	−0.03448943	457	1.575e−02	2.883e−01	VAX2:8 LOXHD1:28 CNTN4:70 PDE6B:74 GABRB3:122 PTEN:151
MP0005464 abnormal platelet physiology	0.08500714	67	1.679e−02	2.960e−01	ABCG5:141 F2RL2:272 P2RY12:336 ITGA2B:409 CLEC1B:416 HPS5:654
MP0002106 abnormal muscle physiology	−0.06135006	128	1.787e−02	3.039e−01	RPTOR:25 DOCK1:38 COL6A1:45 GYS1:68 PLEC:142 SH3BP2:251
MP0000689 abnormal spleen morphology	0.03339451	456	1.950e−02	3.201e−01	MAN2A1:18 CYP1A1:72 HIP1:73 DNMT1:85 RORA:88 MPL:94
MP0002405 respiratory system inflammati	0.06181930	120	2.074e−02	3.290e−01	MAN2A1:18 RORA:88 GPRC5A:413 LTA:428 LEPR:591 PRKDC:671
MP0004043 abnormal pH regulation	0.17720685	12	3.373e−02	3.825e−01	KCNJ1:363 RHCG:518 SCNN1B:756 WNK4:1128 SLC26A7:1283 SLC4A4:2150
MP0002398 abnormal bone marrow	0.02618412	656	3.244e−02	3.825e−01	MAN2A1:18 DST:24 AFF1:31 ANK1:36 HIP1:73 MPL:94
MP0010234 abnormal vibrissa follicle	−0.23940017	7	2.837e−02	3.825e−01	PTEN:151 PKP3:398 CTNNB1:906 CHUK:5124.5 ST14:5124.5 MAP2K1:5124.5
MP0002138 abnormal hepatobiliary system	0.25443077	6	3.099e−02	3.825e−01	PKHD1:521 ABCB4:622 ARG1:1571 MAP3K14:2048 RELB:7610.5 ACVRL1:7610.5
MP0005253 abnormal eye physiology	−0.07681433	69	2.836e−02	3.825e−01	GJA8:141 EGFLAM:202 RGS9BP:209 TRPM1:356 CABP4:454 UCHL3:681
MP0002081 perinatal lethality	−0.02666146	632	3.197e−02	3.825e−01	KIF1A:34 STXBP1:47 BNC2:50 NBEA:51 DYNC1H1:53 TRAF2:55
MP0002080 prenatal lethality	−0.02312318	1023	2.526e−02	3.825e−01	EP400:7 RPTOR:25 DOCK1:38 DYNC1H1:53 ITPR1:54 TRAF2:55
MP0004808 abnormal hematopoietic stem	0.09499002	42	3.385e−02	3.825e−01	MAN2A1:18 HIP1:73 MPL:94 RAPGEF2:398 ITGA2B:409 CSF3R:640
MP0003690 abnormal glial cell	0.11745661	30	2.638e−02	3.825e−01	GALC:214 GRN:241 NPC1:277 PHYH:766 LGI4:1244 CP:1392
MP0002069 abnormal eating/drinking beha	−0.03685555	307	3.111e−02	3.825e−01	DYNC1H1:53 NEB:59 MC4R:92 TSHZ1:110 GABRB3:122 PLEC:142

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
cerebellum	−0.08078509	290	3.379e−06	1.265e−04	POLE:102 MAPK8IP2:109 OTUD7A:114 CDR2L:119 SHANK3:146 APC2:175
liver	0.07909061	294	4.684e−06	1.265e−04	SERPINA7:13 ALB:19 ACY1:35 PBLD:50 IDO2:51 ABCB11:65
testis	0.04191882	1200	7.202e−06	1.296e−04	CFAP20DC:15 FAM187B:25 EFHB:26 MDH1B:40 PKD1L3:42 TIGD4:49
Brodmann.area.9	−0.10372770	123	8.051e−05	9.897e−04	TMEM132D:39 GABRB3:122 CACNG3:159 SYT1:199 KCNQ3:245 SCN1A:332
cerebellar.hemisphere	−0.06072035	363	1.038e−04	9.897e−04	ADCY1:3 CAMK2B:6 VAX2:8 STXBP1:47 NBEA:51 RNF165:85
pancreas	0.12492287	81	1.100e−04	9.897e−04	CPB1:2 CTRC:12 TMED6:28 PNLIP:89 PLA2G1B:146 KCNK16:166
skeletal.muscle	−0.07386356	190	5.220e−04	4.027e−03	CAND2:14 KLHL30:21 NEB:59 MYH2:116 FSD2:123 RGS9BP:209
suprapubic.skin	−0.06621024	174	2.881e−03	1.945e−02	BNC1:96 TMPRSS11F:99 PARD6G:127 KRT80:132 GJB4:193 GSDMA:305
leg.skin	−0.05760348	194	6.278e−03	3.767e−02	TRIM29:81 BNC1:96 TMPRSS11F:99 TREX2:126 PARD6G:127 KRT80:132
cerebral.cortex	−0.08386448	78	1.082e−02	5.070e−02	ADGRB1:27 KIF1A:34 DBNDD1:473 SHC3:514 MCHR1:778 FAM163B:935
cortex.kidney	0.07356735	100	1.152e−02	5.070e−02	SLC5A12:5 CUBN:20 CLDN16:66 TMEM52B:108 ENAM:113 SLC6A12:341
EBV.lymphocyte	0.03259584	576	1.006e−02	5.070e−02	UHRF1:29 TMEM131L:63 SPDL1:69 CRYZ:70 ACADM:92 ATAD5:117
thyroid	0.06515877	126	1.220e−02	5.070e−02	PLA2R1:43 PRTG:44 SLC12A8:45 IDO2:51 PRSS45P:57 TLE6:121
hypothalamus	−0.10461345	45	1.546e−02	5.962e−02	MC4R:92 PIRT:194 SYT6:611 OTP:688 GRIK1:876 DNER:1039
tibial.artery	−0.07359460	86	1.894e−02	6.819e−02	SDK1:80 TLN1:94 LMOD1:267 ACTA2:296 CDH6:328 C1QTNF1:519
breast	−0.18835519	12	2.396e−02	7.551e−02	PLIN1:367 TFAP2B:644 THRSP:849 ABCB5:5124.5 ADIPOQ:5124.5 ALX4:5124.5
esophagus.mucosa	−0.04960156	175	2.517e−02	7.551e−02	BNC1:96 TMPRSS11F:99 FZD10:198 PGLYRP4:310 PKP3:398 TGM1:410
peyers.patch	0.05341449	152	2.436e−02	7.551e−02	SLC5A12:5 MALRD1:56 SLC23A1:68 SLC13A2:118 GUCA2B:123 MEP1A:171
C1.spinal.cord	−0.05308646	134	3.536e−02	9.880e−02	PDE6B:74 ABCA2:178 MBP:206 SEMA4D:240 NKAIN2:275 ZNF536:439
fallopian.tube	0.09131778	44	3.659e−02	9.880e−02	GMNC:165 ADAMTS3:278 ADGRG4:512 CRISPLD2:944 CFAP157:1000 CDHR4:1030
minor.salivary.gland	0.07427224	62	4.390e−02	1.092e−01	BPIFB1:1 ENPP3:192 ASCL3:393 SCNN1A:442 PROM1:477 GALNT3:554
eye.development	−0.09566717	37	4.450e−02	1.092e−01	NRL:257 OTX2:557 NOTCH1:672 FGFR2:5124.5 PAX6:5124.5 LHX2:5124.5
nucleus.accumbens	−0.07160809	64	4.844e−02	1.137e−01	JAKMIP1:1 SLC32A1:218 SORCS3:278 INSYN2A:633 OPRK1:694 KCTD4:805
stomach	0.06332206	74	6.081e−02	1.368e−01	PGC:3 ADAM28:14 TMED6:28 TMPRSS2:145 CA9:290 SLC9A4:403
blood	0.03191640	235	9.681e−02	2.028e−01	LY6G6F:52 TREML2:54 RHAG:185 SLC4A1:221 AQP9:228 SLC6A6:256
sigmoid.colon	−0.07557609	39	1.032e−01	2.028e−01	TACR2:20 HDAC4:118 CNN1:397 NTSR1:479 CHRM2:800 HAND2:847
transformed.skin.fibroblast	−0.03256528	221	9.995e−02	2.028e−01	COL6A1:45 CNTN3:88 BNC1:96 OPN3:105 TENM3:135 HAS2:152
vagina	−0.05400923	76	1.052e−01	2.028e−01	BNC1:96 TMPRSS11F:99 PGLYRP4:310 PKP1:465 CNFN:481 DSG3:540
spleen	0.02751983	261	1.322e−01	2.380e−01	AKNA:23 CLEC4F:90 DENND1C:93 MPEG1:144 BANK1:264 FGD2:298
transverse.colon	0.03899734	128	1.307e−01	2.380e−01	TRPM6:4 GUCA2B:123 ADAMDEC1:301 MUC4:337 TMIGD1:355 SLC26A2:495
Brodmann.area.24	−0.09004728	21	1.536e−01	2.568e−01	STUM:241 FAM189A1:689 NEUROD6:1118 CNTN5:5124.5 CRH:5124.5 CSPG5:5124.5
endocervix	−0.05450561	59	1.491e−01	2.568e−01	ZNF516:101 SPON2:148 TMEM132C:166 KIF26A:414 VIPR2:423 NXNL2:537
putamen	−0.12933765	10	1.569e−01	2.568e−01	RGS9:950 DIPK1C:1146 ANO3:5124.5 FAM237A:5124.5 SYNDIG1L:5124.5 KCNH4:5124.5
esophagogastric.junction	0.17955087	5	1.645e−01	2.613e−01	F2RL2:272 GADL1:1330 COL4A5:3214 ADCY5:7610.5 CHRM3:7610.5 NA
amygdala	−0.14696587	7	1.783e−01	2.751e−01	TNR:951 NEUROD6:1118 LHFPL3:5124.5 PCDH15:5124.5 PTPRZ1:5124.5 SLC38A8:5124.5
pituitary.gland	−0.02541057	170	2.581e−01	3.871e−01	GAL:40 RIMBP2:128 NSG1:167 BAIAP3:176 CRHR2:186 IGSF1:223
lung	0.02955347	120	2.670e−01	3.897e−01	ADGRF5:328 GPRC5A:413 A2M:550 CSF3R:640 ADGRL2:727 GP9:825
ectocervix	0.05950738	22	3.346e−01	4.754e−01	RIPOR3:203 COL27A1:271 ADRA2A:732 MMRN1:977 PTHLH:1692 PAPPA2:2098
substantia.nigra	−0.08195130	10	3.698e−01	5.120e−01	TH:471 CHRM5:5124.5 FOXB1:5124.5 C10orf105:5124.5 KLHL1:5124.5 RET:5124.5
caudate.nucleus	−0.05064802	19	4.452e−01	5.724e−01	MLC1:813 PDE1B:991 ANO3:5124.5 CHAT:5124.5 FAM237A:5124.5 GPR37L1:5124.5